

Alessandro Trinca, Ph.D.

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 @ORCID



Main Research Interests

- Black hole formation and growth across cosmic time
- Active Galactic Nuclei populations at high-redshift
- Dynamics and evolution of massive black hole binaries as GW sources
- Early galaxy formation and its connection to massive black holes
- Properties of the first stellar populations

Employment

- 2025 – Now  **Postdoctoral Researcher** - Institute for Astronomy, University of Edinburgh,
Project "*Black hole dynamics and the origin of supermassive black holes*"
PI: Dr. Ricarda Beckmann
- 2024 – 2025  **Postdoctoral Researcher** - University of Insubria,
Project "*Formation and evolution of massive black holes in quasar host galaxies at high redshift*"
PI: Prof. Alessandro Lupi
- 2023 – 2024  **Postdoctoral Researcher** - INAF Astronomical Observatory of Rome,
Project "*Investigating early black holes growth at cosmic dawn with light and gravitational waves*"
PI: Dr. Rosa Valiante

Education

- 2019 – 2023  **Ph.D. in Astronomy, Astrophysics and Space Science**, University of Rome "La Sapienza" & University of Rome "Tor Vergata".
Thesis: "*Light and sound from the cosmic dawn: the origin and observational signatures of the first black holes*",
PhD Advisor: Prof. Raffaella Schneider
- 2017 – 2019  **Master's Degree in Astronomy and Astrophysics, *cum laude***, University of Rome "La Sapienza"
Thesis: "*The low mass end of the black hole mass function at cosmic dawn*",
Supervisor: Prof. Raffaella Schneider,
- 2013 – 2017  **Bachelor's Degree in Physics and Astrophysics**, University of Rome "La Sapienza"
Thesis: "*The formation of the first supermassive black holes*"
Supervisor: Prof. Raffaella Schneider,

Grants & Funding

Grants

- 2023  **INAF Mini-Grant – Bando di Ricerca Fondamentale 2023**
PI of the project: *Cosmic Archaeology with the first black hole seeds: advancing theoretical tools to explore cosmic dawn with JWST and future gravitational wave telescopes*
Awarded **20.000 EUR** for 2 years

Computational & Observational Programs

- 2025  **EuroHPC Computing Grant**, Development Access (0.5 MCPUh) **PI**, Title: *The Rise of Cosmic Giants: exploring the birthplaces of the first massive black holes*
-  **JWST DD Cycle 4** (3.7 hrs), STScI, **Co-I**, PI: Maiolino
Title: *Black hole primacy - do supermassive black holes really predate galaxies?*
-  **JWST GO Cycle 4** (62.8 hrs), STScI, **Co-I**, PI: Calabró
Title: *Searching for Population III stars around massive interacting galaxies in the reionization epoch*
- 2024  **JWST Large Program**, Cycle 3 (137.82 hrs), STScI, **Co-I**, PI: Übler
Title: *Unveiling the nature and impact of the first population of black holes: an extensive NIRSpec-IFU survey in the first billion years*
-  **JWST GO Cycle 3** (15.29 hrs), STScI, **Co-I**, PI: Fujimoto
Title: *Panchromatic characterizations of the super-Eddington accretion black hole, host, and environment: Epicenter of red dots, mergers, and dusty starbursts at $z=7.2$*
- 2023  **JWST GO Cycle 2** (46.28 hrs), STScI, **Co-I**, PI: Übler
Title: *The search for Population III stars in low-metallicity $7 < z < 9.5$ galaxies*

Scholarship

- 2019 – 2022  **Doctoral Scholarship at University of Rome “Tor Vergata”**
Joint PhD Program in Astronomy, Astrophysics and Space Science University of Rome “Tor Vergata” / University of Rome “La Sapienza”

Service

Student Supervision

Master

- 2023 – 2024  Miriana Bernardi, Università degli Studi di Roma “La Sapienza”
Thesis: *"Dynamical evolution of cosmological black hole pairs: imprints on the observability of the first black hole population"*
- 2022 – 2023  Arianna Ferrotti, Università degli Studi di Roma “La Sapienza”
Thesis: *"Observational imprints of a non-universal stellar initial mass function"*
- 2021  Emanuele Ventura, Università degli Studi di Padova
Thesis: *"Modeling the 21cm global signal from first stars and black holes"*

Service (continued)

Bachelor

- 2025  Yari Spoleto Albonico, Università degli Studi dell'Insubria
Thesis: *"Detectability of coalescing binary black holes at high redshift with the Einstein Telescope."*
-  Giulia Liguori, Università degli Studi dell'Insubria
Thesis: *"The impact of gravitational kicks on the ejection of massive black holes from their host galaxies at high redshift."*

Peer Reviewer

- 2023 – Now  Monthly Notices of the Royal Astronomical Society (MNRAS)
-  Astronomy & Astrophysics (A&A)
- 2024 – Now  Open Journal of Astrophysics (OJAp)
-  Astrophysical Journal (ApJ)
- 2026–present  Nature Communications (Nat. Commun.)

Organizing Committee

- 2021 – 2022  *"QUID ULTRA?" Frontiers and Controversies in Astrophysics,*
Series of Monthly Colloquia, Sapienza University of Rome

Collaborations & Memberships

- 2022 - Now  **LISA Consortium**, Core Member - Project Coordinator within the Astrophysics Working Group
-  **Eistein Telescope Collaboration**, Division 3 - Population Studies
- 2019 - Now  **INAF - Istituto Nazionale di Astrofisica**
-  **INFN - Istituto Nazionale di Fisica Nucleare**

Contributions and Seminars

Invited Talks/Seminars

- 2026  Computational Galaxy Formation Workshop
Ringberg Castle, Germany
-  Cosmic Scene Investigation in Sexten
Sexten Center for Astrophysics, Sexten, Italy
-  The growth of galaxies in the early Universe XI
Sexten Center for Astrophysics, Sexten, Italy

Contributions and Seminars (continued)

- 2025  Departmental Seminar
Department of Science and Advanced Technology, University of Insubria, Como Italy
- 2025  Astrophysics Seminar
University of Milano-Bicocca, Milan, Italy
- 2024  The Slippery Slope of Multimessenger Cosmological Black Holes Data
Sexten Center for Astrophysics, Sexten, Italy
- 2024  Cosmic Dawn Revealed by JWST: The Physics of the First Stars, Galaxies, and Black Holes
Kavli Institute for Theoretical Physics, University of California, Santa Barbara
- 2023  Astrophysics Seminar
Institute for Computational Science, Zurich, Switzerland.

Contributed Conference Talks

- 2026  Breaking the Limits: Super-Eddington Accretion onto Compact Objects
Cagliari, Italy
-  Active Galactic Nuclei XVI
Brindisi, Italy
-  Massive Black Hole Spin Workshop
The Higgs Centre for Theoretical Physics, Edinburgh, UK
-  22nd Durham-Edinburgh eXtragalactic Workshop
University of Edinburgh, Edinburgh, UK
- 2025  Massive Black Holes across Cosmic Time
Kavli Institute for Cosmology, Cambridge, UK
-  EREBUS / COSMOS-3D collaboration meeting
Bologna, Italy
-  CRISOL25 - Galaxy Origins in the JWST Era
Toledo, Spain
- 2024  40th IAP Symposium
Institut Astrophysique de Paris, France
-  LISA Astrophysics Working Group 2024
Max Planck Institute for Astrophysics, Garching, Germany.
-  Young Astronomers on Galactic Nuclei 2024
University of Insubria, Como, Italy.
-  The Origin and Evolution of Supermassive Black Holes
Sexten Center for Astrophysics – Sexten, Italy.
-  First Stars VII Conference
Center for Computational Astrophysics – New York, US.
-  Massive Black Holes in the First Billion Years
Kinsale, Ireland
- 2023  Black Holes on Broadway Conference
Center for Computational Astrophysics – New York, US.
-  Young Astronomers on Galactic Nuclei 2023
INAF-IASF, Palermo, Italy.
-  LISA Astro Working Group Meeting
Bicocca University, Milan, Italy.
-  Fake Light Conference
Center for Computational Astrophysics – New York, US.

Contributions and Seminars (continued)

- European Astronomical Society Meeting 2023
Krakow, Poland.
- HSC Meeting 2023: HYPERION, WISSH & Friends
Bologna, Italy.
- 2022 ■ "Origin, growth and feedback of black holes in dwarf galaxies"
DIPC, San Sebastian, Spain.
- European Astronomical Society Meeting 2022
Valencia, Spain.
- AGN XIV: The Renaissance of Black Holes and Galaxies
Florence, Italy.
- 2020 ■ Young Astronomers on Galactic Nuclei 2020
Niels Bohr Institute, Denmark.

Doctoral Schools and Workshops

- 2022 ■ DAWN Winter School
Cosmic Dawn Center, Niels Bohr Institute, Copenhagen, Denmark
- Saas-Fee Advanced Course
"Compact-Object Astrophysics in the Era of Multi-Messenger Astronomy"
Saas-Fee, Switzerland
- 2020 ■ Introduction to Parallel Computing with MPI and OpenMP
CINECA Academy, Bologna, Italy
- LISA Astrophysics Working Group Workshop
Nijmegen, Netherlands

Skills

- Languages ■ ITALIAN (Mother tongue), ENGLISH (Professional working proficiency), FRENCH (Elementary proficiency)
- Computing ■
 - *Programming Languages*: Fortran, C, C++, Python, R
 - *Scientific Libraries*: NumPy, SciPy, Matplotlib, pandas, Astropy, yt, h5py
 - *Parallel Computing*: MPI, OpenMP
 - *Development Tools*: Git, Jupyter, Conda,
 - *Others*: SLURM, HTML, CSS

Academic References

Prof. Raffaella Schneider

University of Rome "La Sapienza",
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Dr. Rosa Valiante

INAF - Astronomical Observatory of Rome,
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Prof. Alessandro Lupi

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alessandro.lupi@uninsubria.it